



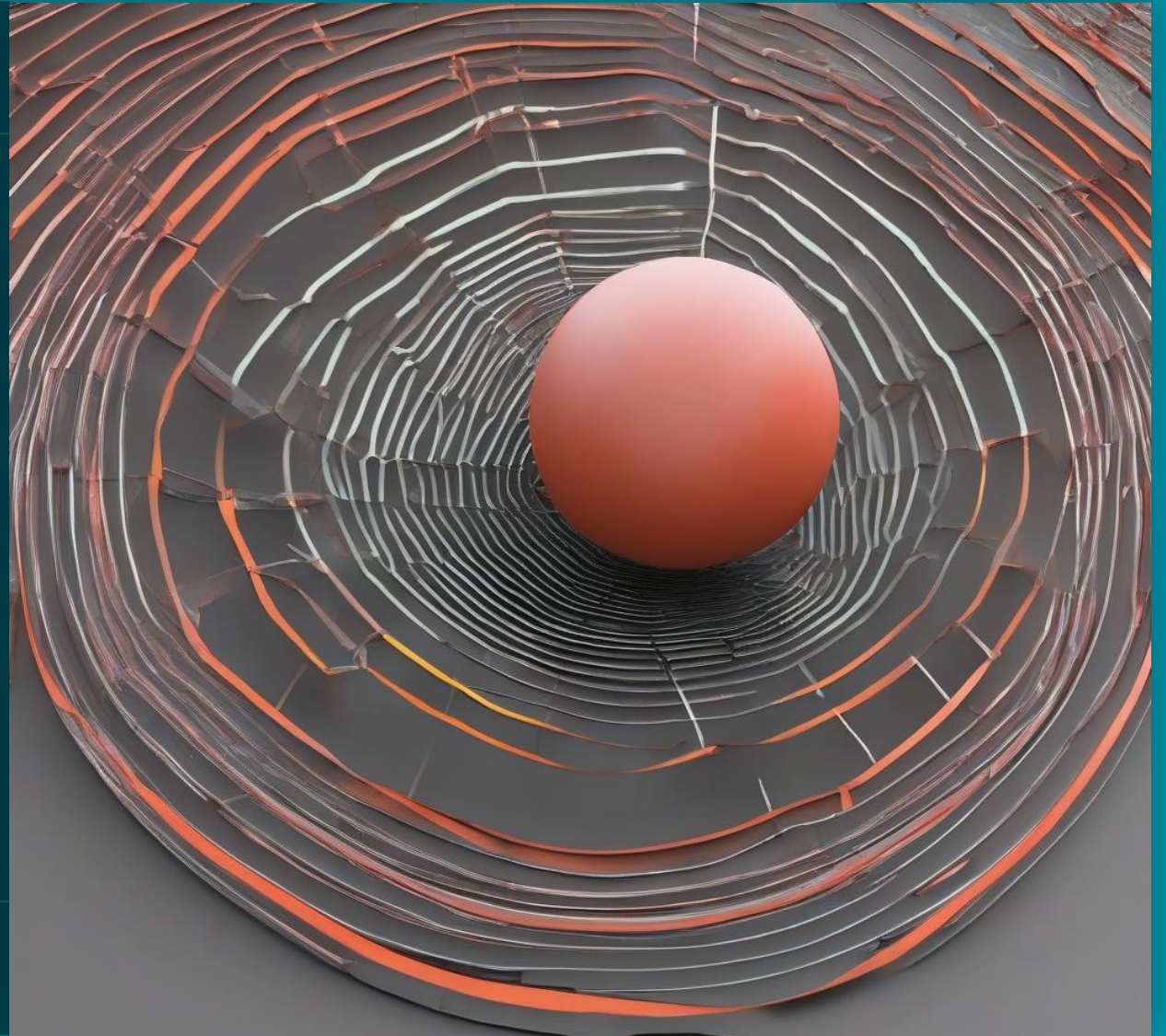
HOSPITAL DA LUZ
LEARNING HEALTH+

Introduction to
Machine Learning in
Healthcare

Model Evaluation and Optimization

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Model Tuning and Optimization

RELEVANCE TO MEDICINE

Resource Allocation



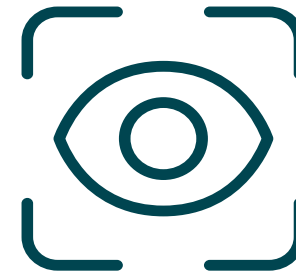
Patient Safety



Accuracy and Reliability of Diagnoses



Personalized Medicine



KEY TERMS AND CONCEPTS

- Classification
- Regression
- accuracy, precision, recall, F1 score, RMSE, MAE, etc.
- Train-Test Split
- Hyperparameters

The background of the entire slide is a dense crowd of stylized human figures in red, blue, and white. The figures are out of focus, creating a bokeh effect. On the left side, there is a vertical teal bar that serves as a background for the title and number.

01

MODEL EVALUATION METRICS FOR CLASSIFICATION

INTRODUCTION TO MODEL EVALUATION METRICS

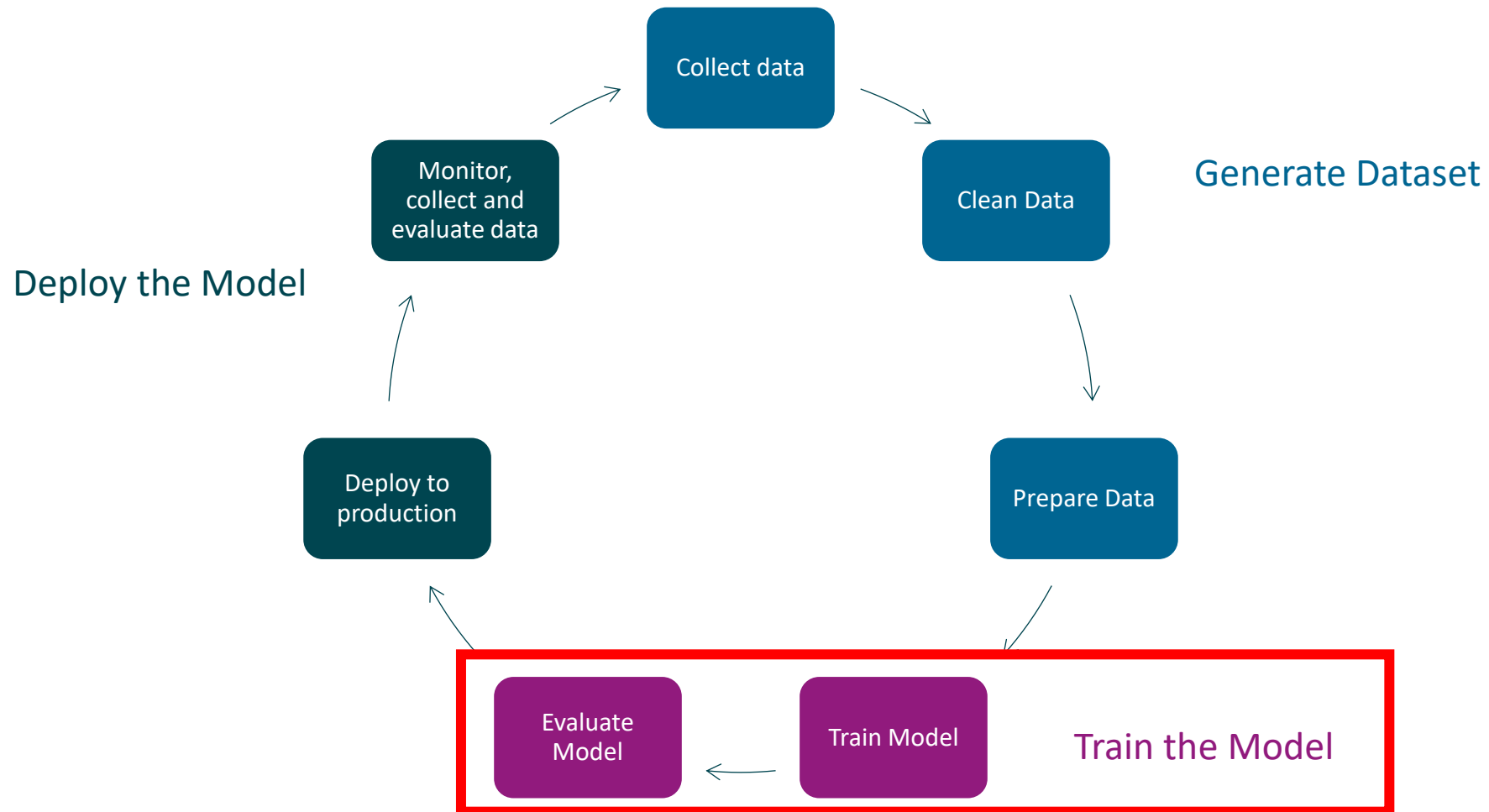
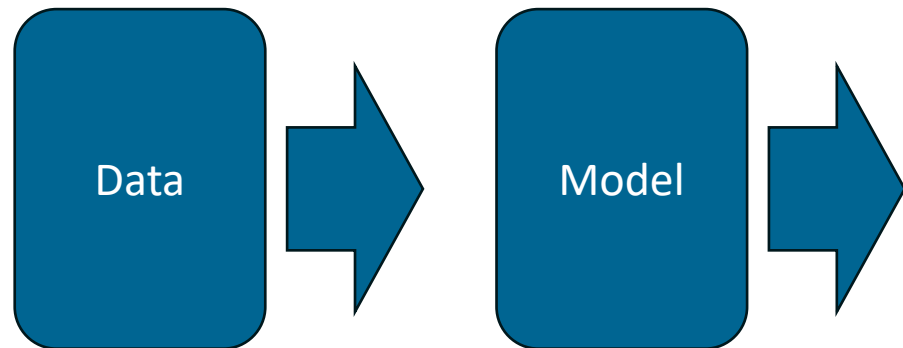


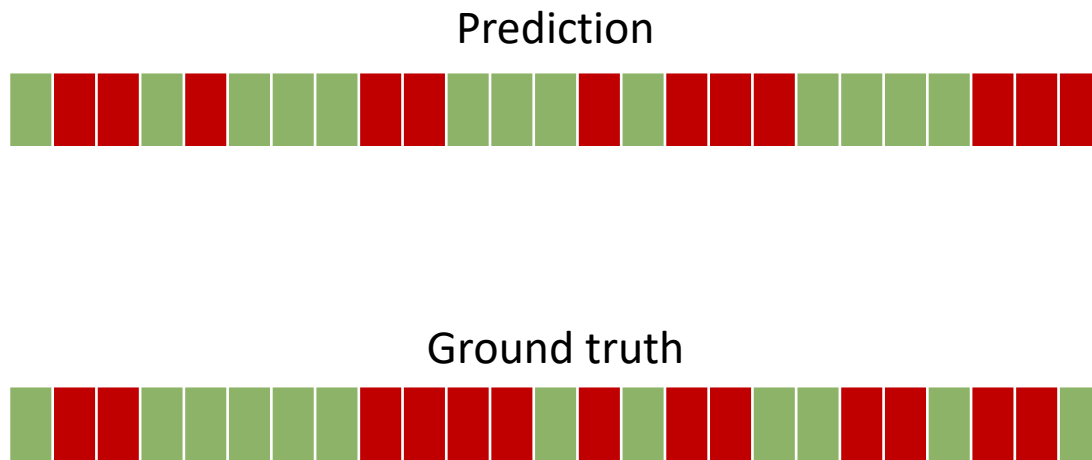
Image from: [What is a Machine Learning Pipeline? - Datatron](#)

MODEL OUTPUT

Classification model



CONFUSION MATRIX



		True Class	
		Positive	Negative
Predicated Class	Positive	9	3
	Negative	4	9

ACCURACY

measures the **proportion of true results** (both true positives and true negatives) among the total number of cases examined

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

ACCURACY

Main limitation

Class imbalance

		True Class	
		Positive	Negative
Predicated Class	Positive	95	5
	Negative	0	0

$$Accuracy = \frac{95 + 0}{95 + 0 + 5 + 0} = 0.95$$

CONFUSION MATRIX

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

PRECISION

measures the accuracy of the positive predictions

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

PRECISION

Main limitation

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Precision does not consider the false negatives, so it does not provide information about the ability of the model to identify all actual positives

PRECISION

Case study

Case Study Example

Consider a model designed to predict whether patients have a particular rare genetic disorder. High precision is crucial because:

- False positives could lead to expensive and invasive follow-up tests.
- Ensuring those diagnosed are highly likely to have the disorder can help in focusing medical resources effectively.

RECALL (SENSITIVITY)

quantifies the ability of the model to find all relevant cases within a dataset

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

RECALL (SENSITIVITY)

Main limitation

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Recall does not consider the false positives, so it doesn't provide information about the accuracy of the positive predictions

RECALL (SENSITIVITY)

Case study

Case Study Example

Consider a model designed to predict whether patients have a highly infectious disease. High recall is crucial because:

- Missing actual cases (false negatives) could result in the spread of the disease.
- Early identification of all infected individuals allows for prompt isolation and treatment to prevent further transmission.

F1 SCORE

is the harmonic mean of
precision and recall

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 SCORE

Main limitation

Interpretability

Class Imbalance Sensitivity

SPECIFICITY

True Negative Rate (TNR)

quantifies the ability of the model to
correctly identify negative cases

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

SPECIFICITY

Main limitation

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Specificity does not consider the false negatives, so it doesn't provide information about the model's ability to identify all actual positive instances

SPECIFICITY

True Negative Rate (TNR)

Case Study Example

Consider a model designed to screen for a non-life-threatening condition that has a simple but expensive treatment. High specificity is crucial because:

- False positives could lead to unnecessary and costly treatments.
- Ensuring that patients who do not need the treatment are not incorrectly identified as needing it helps in managing healthcare costs and resources effectively.

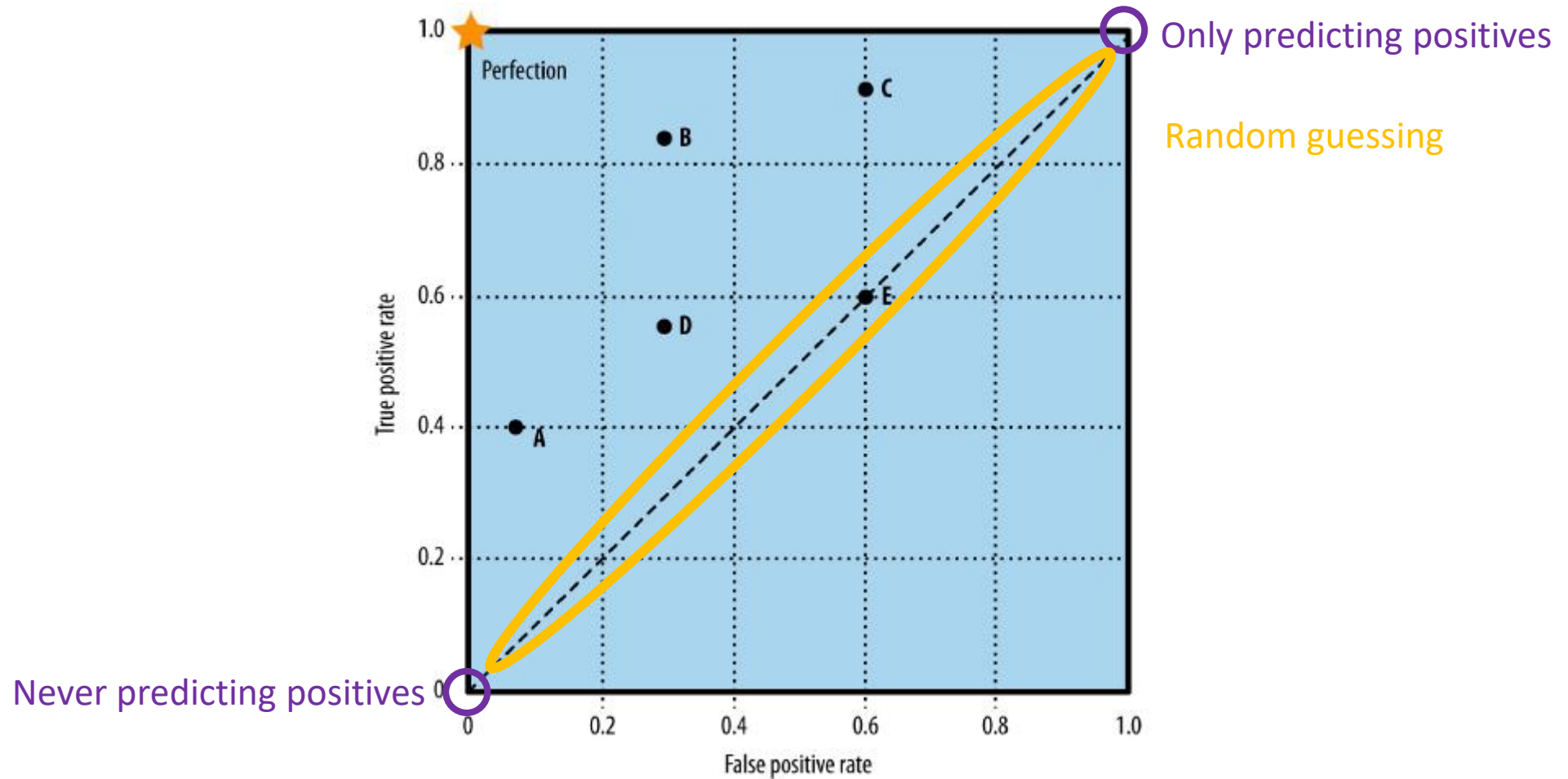
ROC CURVE

used to evaluate the performance of a **binary classification** model

trade-off between **sensitivity** (true positive rate) and **1-specificity** (false positive rate) at **various threshold** settings

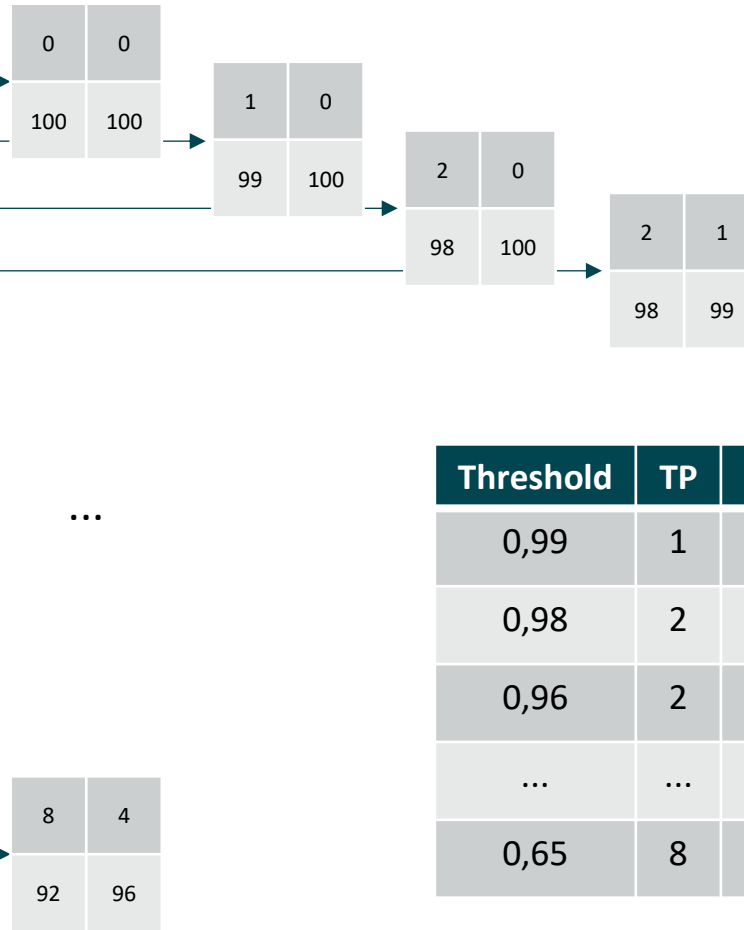
helps in **visualizing** the ability of the classifier to distinguish between positive and negative instances

ROC CURVE



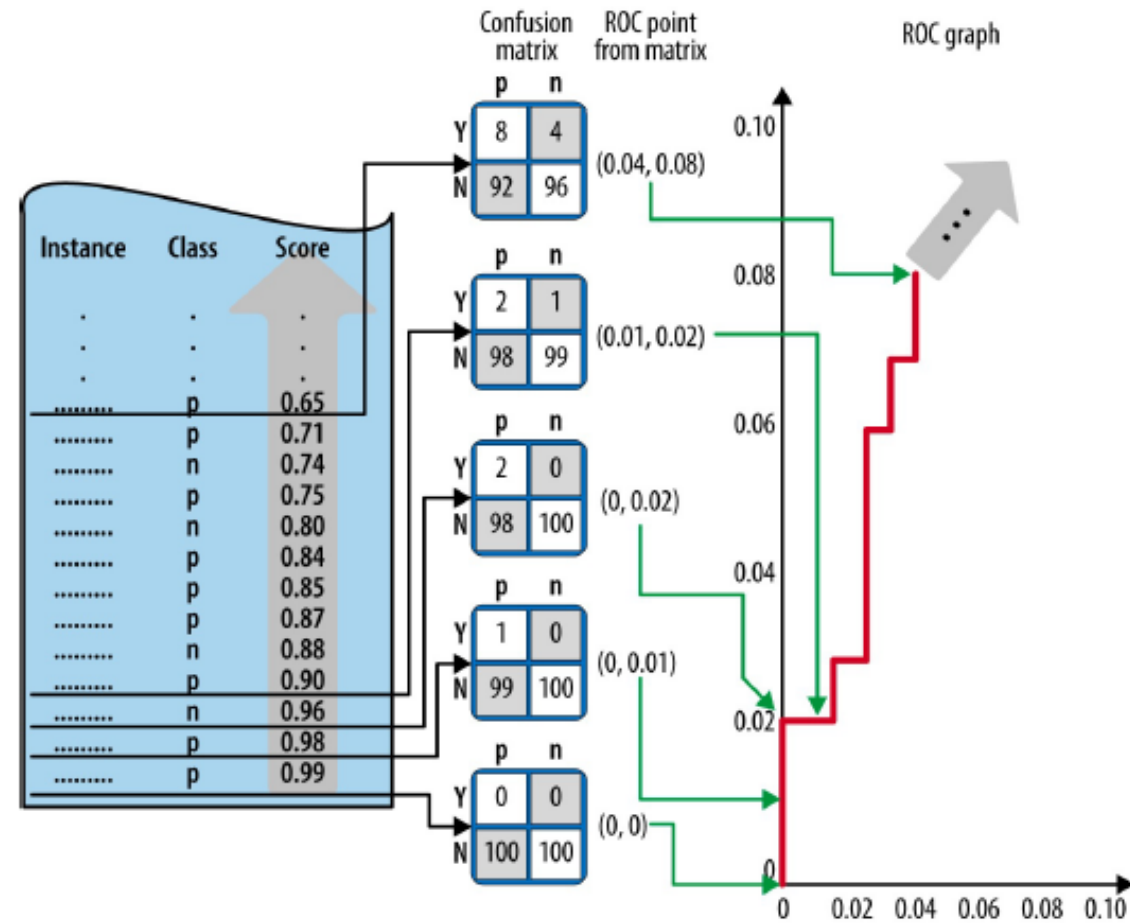
ROC CURVE

Instance	True Label	Prediction Score
1	P	0,99
2	P	0,98
3	N	0,96
4	P	0,90
5	N	0,88
6	P	0,87
7	P	0,85
8	P	0,84
9	N	0,80
10	P	0,75
11	N	0,74
12	P	0,71
13	P	0,65

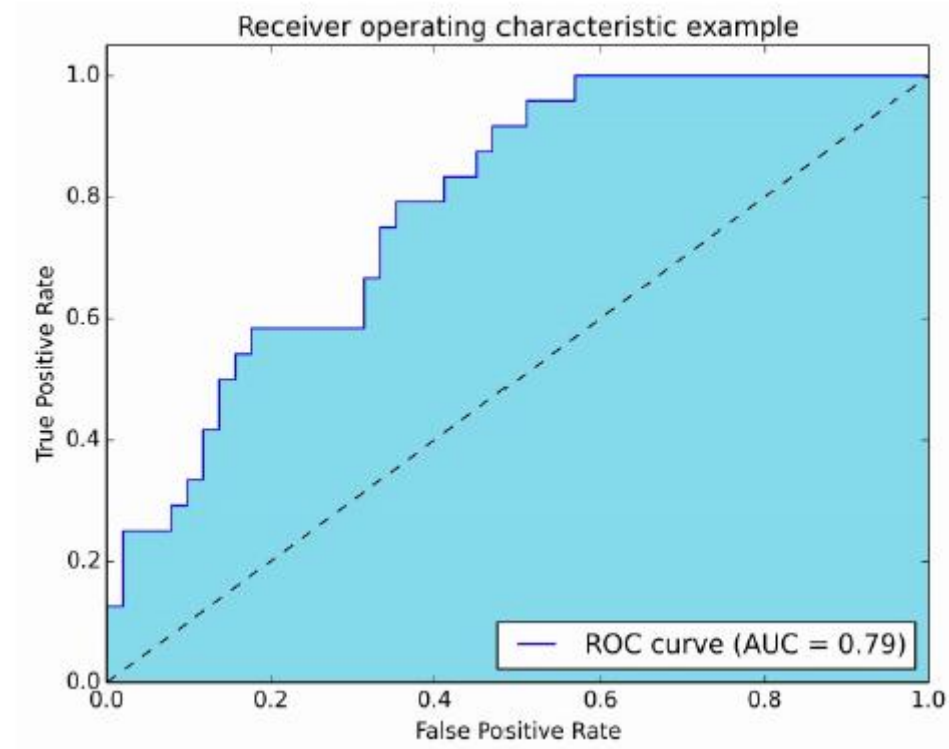


Threshold	TP	FP	TN	FN	TPR	FPR
0,99	1	0	100	99	0,01	0
0,98	2	0	100	98	0,02	0
0,96	2	1	99	98	0,02	0,01
...	...	...	...	...	...	...
0,65	8	4	96	92	0,04	0,08

ROC CURVE



ROC CURVE



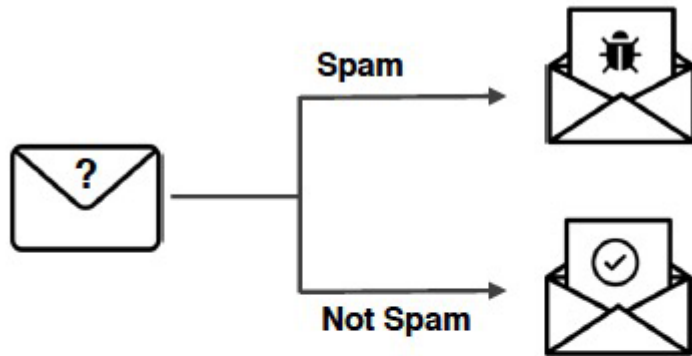
<http://www.navan.name/roc/>

AUC (AREA UNDER THE ROC CURVE)

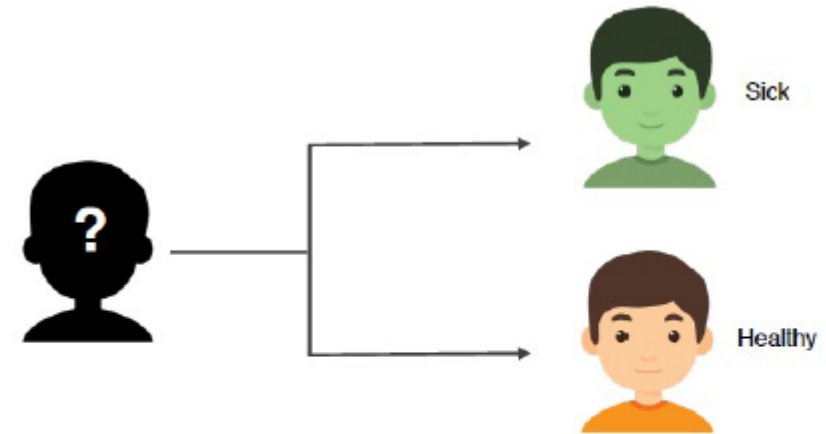
Practical Application in Medicine

- **Evaluating Diagnostic Tests:** A higher AUC indicates better performance in distinguishing between patients with and without a condition.
- **Model Comparison**
- **Threshold Selection**

DISCUSSION



- Higher false negative rate means there might be more junk mails. Although more junk emails are annoying, it's better than missing the important email.



- Early diagnosis of malignant tumor is very important. Not finding malignant tumor is worse than misclassifying benign tumor as malignant one.

DISCUSSION

Rastreios vs tratamento ao cancro
Recall vs recall

DECISION MAKING

original reports

Developing a Prediction Model for Pathologic Complete Response Following Neoadjuvant Chemotherapy in Breast Cancer: A Comparison of Model Building Approaches

Robert B. Basmadjian, MSc¹; Shiyong Kong, MSc^{1,2,3}; Devon J. Boyne, PhD^{1,2}; Tamer N. Jarada, PhD²; Yuan Xu, MD, PhD^{1,2,3}; Winson Y. Cheung, MD^{1,2}; Sasha Lupichuk, MD^{1,2}; May Lynn Quan, MD^{1,2,3}; and Darren R. Brenner, PhD^{1,2}

<https://doi.org/10.1200/CCI.21.00055>

EXERCISE

- notebook

METRIC SELECTION

Some considerations when selecting the right metric for the classification task:

- Binary vs. Multiclass Classification
- Imbalanced Classes
- Objective of the Model (minimizing false positives or negatives)
- Threshold-Independent or -Dependent Metrics
- Interpretability
- Clinical Utility

02

MODEL EVALUATION METRICS FOR REGRESSION



INTRODUCTION TO MODEL EVALUATION METRICS

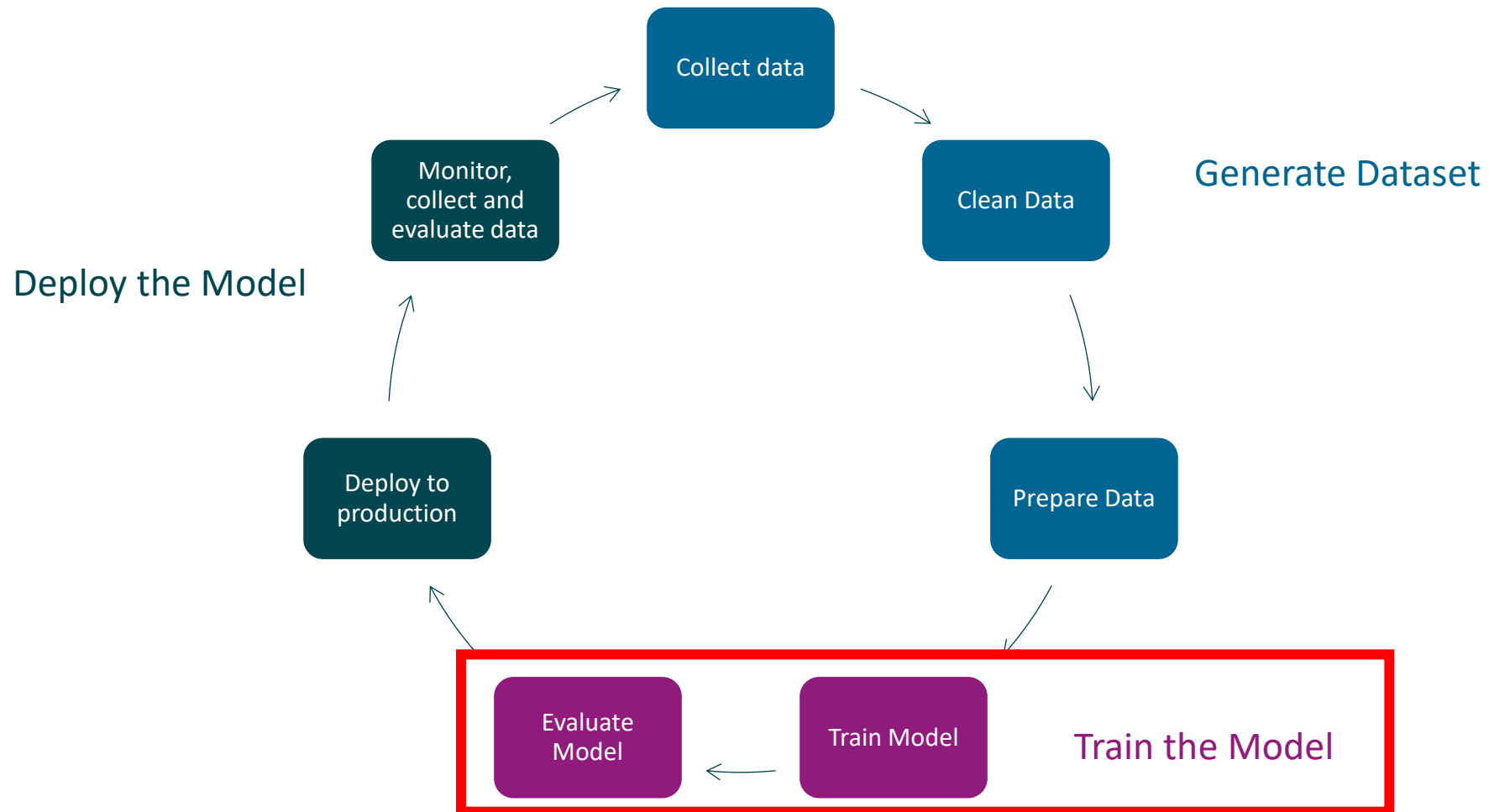
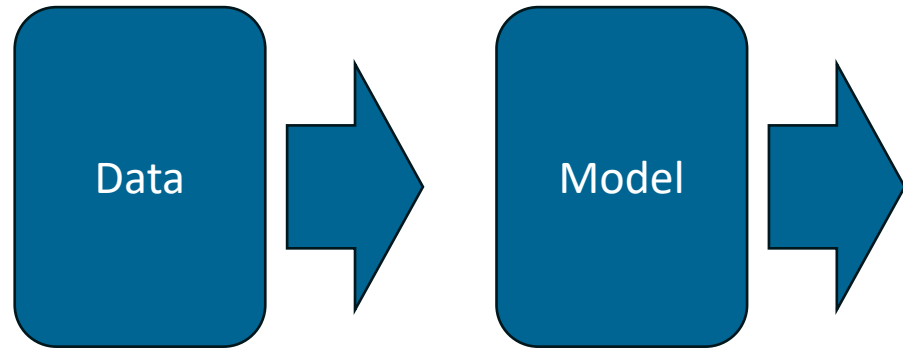


Image from: [What is a Machine Learning Pipeline? - Datatron](#)

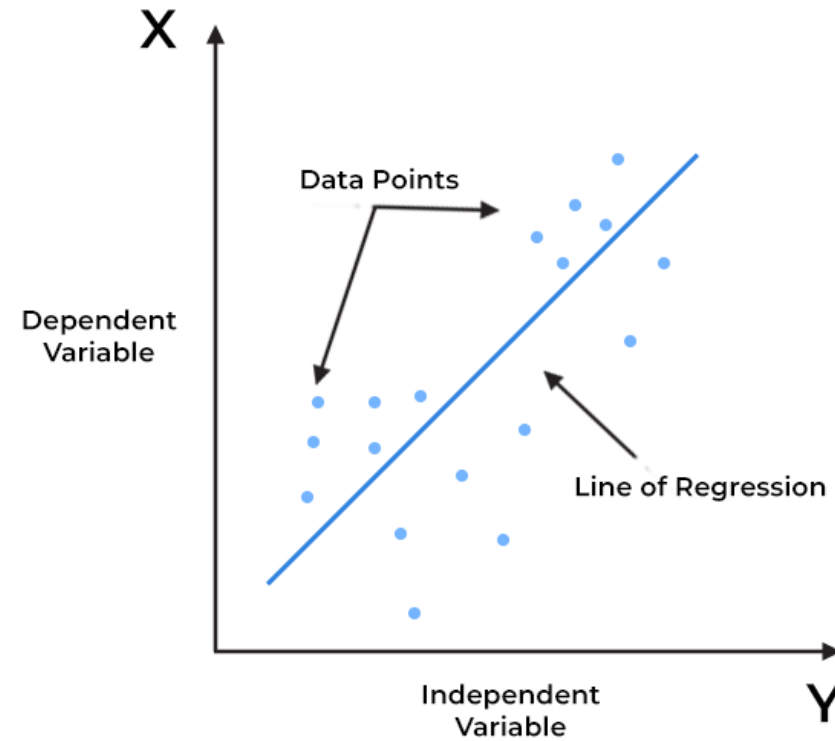
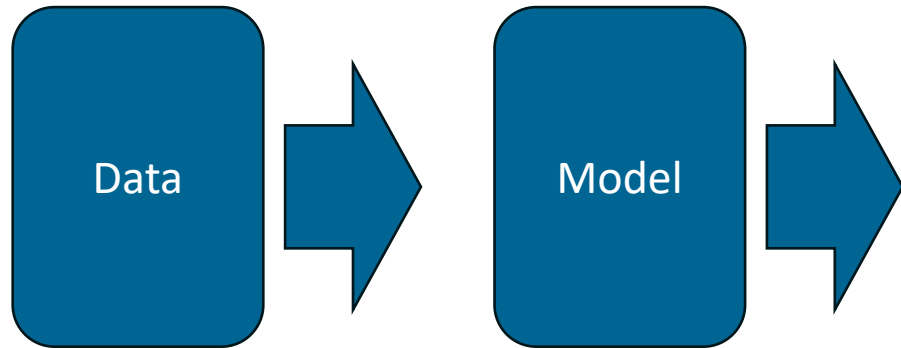
MODEL OUTPUT

Regression model



MODEL OUTPUT

Regression model



MODEL OUTPUT

Metrics

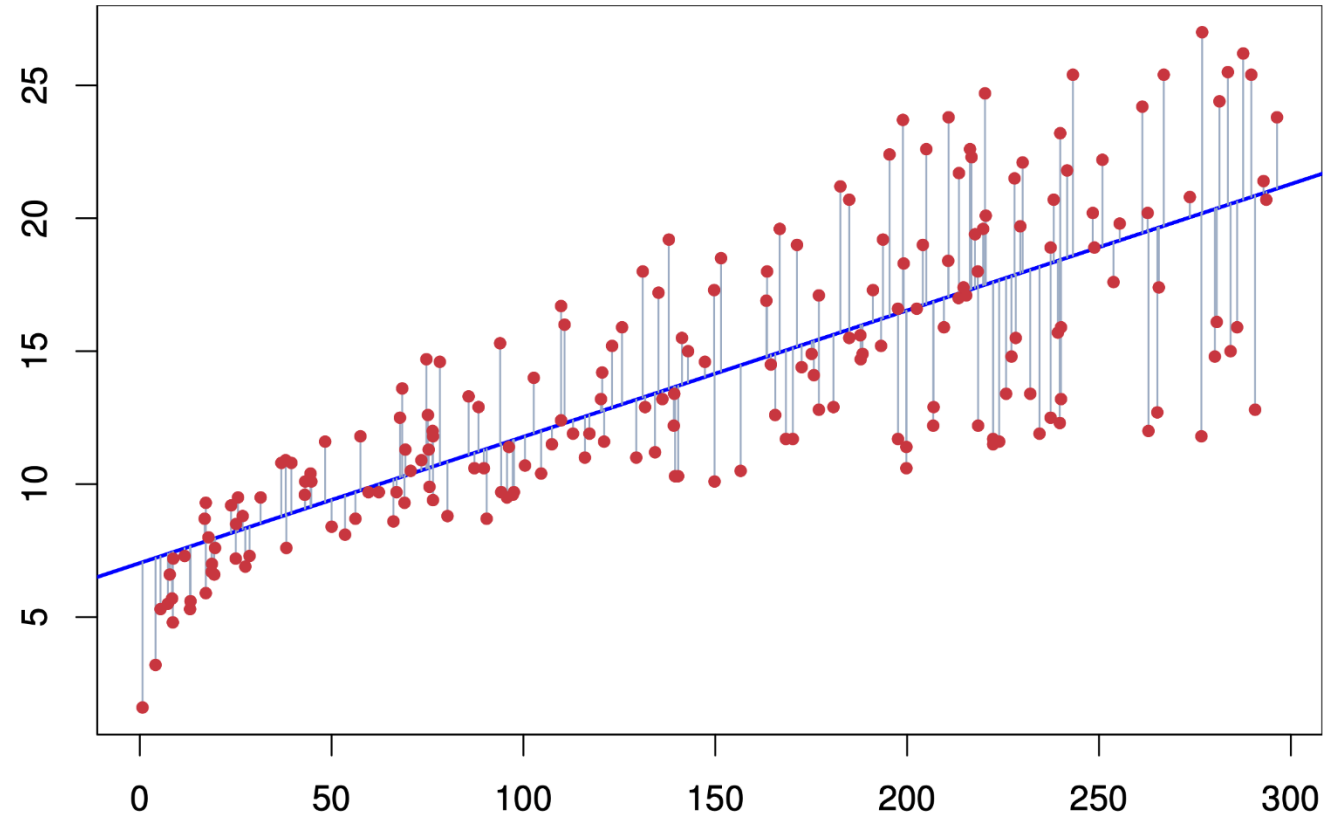
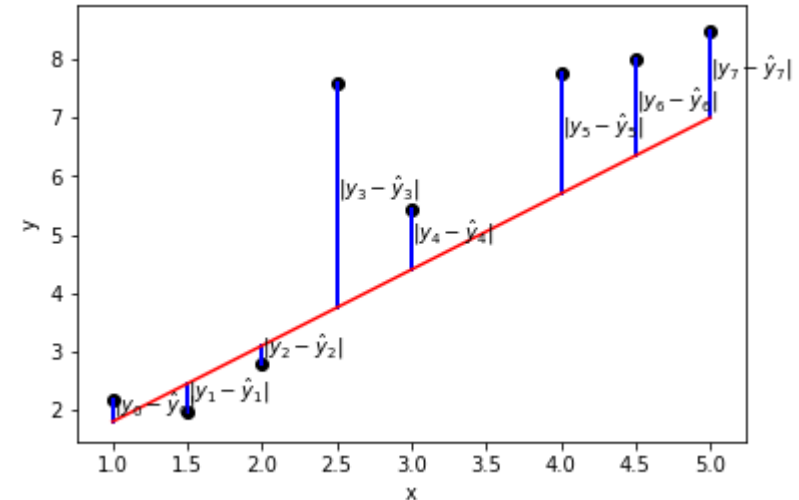


Image from: <https://web.stanford.edu/class/stats202/notes/Linear-regression/Simple-linear-regression.html>

MEAN ABSOLUTE ERROR (MAE)

measures the average magnitude of errors
between predicted and actual values

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$



MEAN SQUARED ERROR (MSE)

Importance

- **Interpretability:** MAE is easy to understand and interpret, as it represents the average error in the same units as the target variable.
- **Robustness to Outliers:** Unlike metrics such as Mean Squared Error (MSE), MAE is less sensitive to outliers because it does not square the error terms.

Limitations

- **Equal Weight to All Errors:** MAE gives equal weight to all errors, regardless of their magnitude. This can be a limitation in cases where larger errors need to be penalized more.

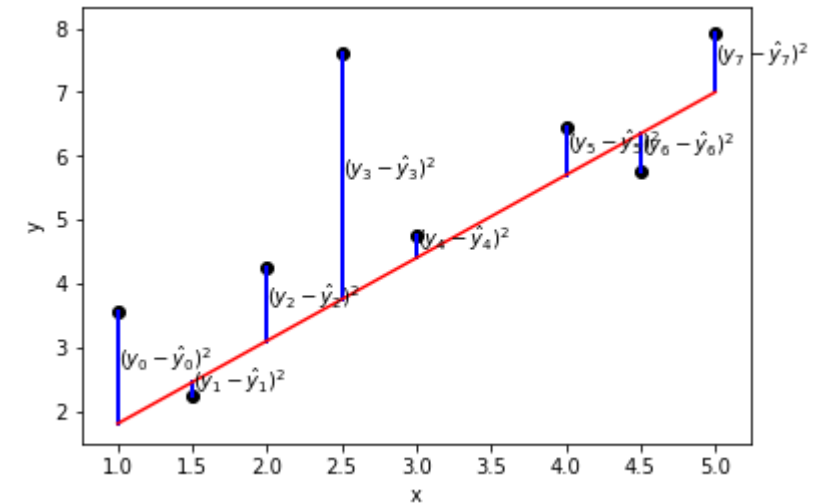
EXERCISE

- $Y = 1, 3, 5$
- $Y_{\text{pred}} = 0.5, 3, 5$

ROOT MEAN SQUARED ERROR (RMSE)

measures the square root of the average
of the squared differences between
predicted and actual values

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



ROOT MEAN SQUARED ERROR (RMSE)

Importance

- **Sensitive to Large Errors:** RMSE is more **sensitive to large errors** compared to MAE because the errors are squared before averaging. This means that RMSE can be particularly useful when large errors are undesirable.

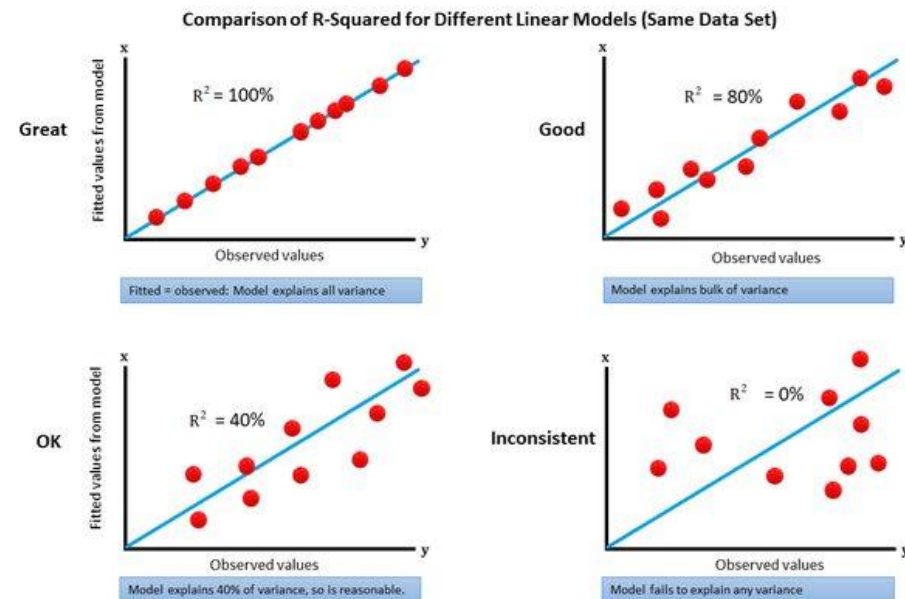
Limitations

- **Sensitivity to Outliers:** Because RMSE squares the error terms, it can be **disproportionately influenced by outliers**. Large errors will have a greater impact on RMSE.
- **Comparability:** RMSE values are context-specific and can only be used to compare models on the same dataset or problem.

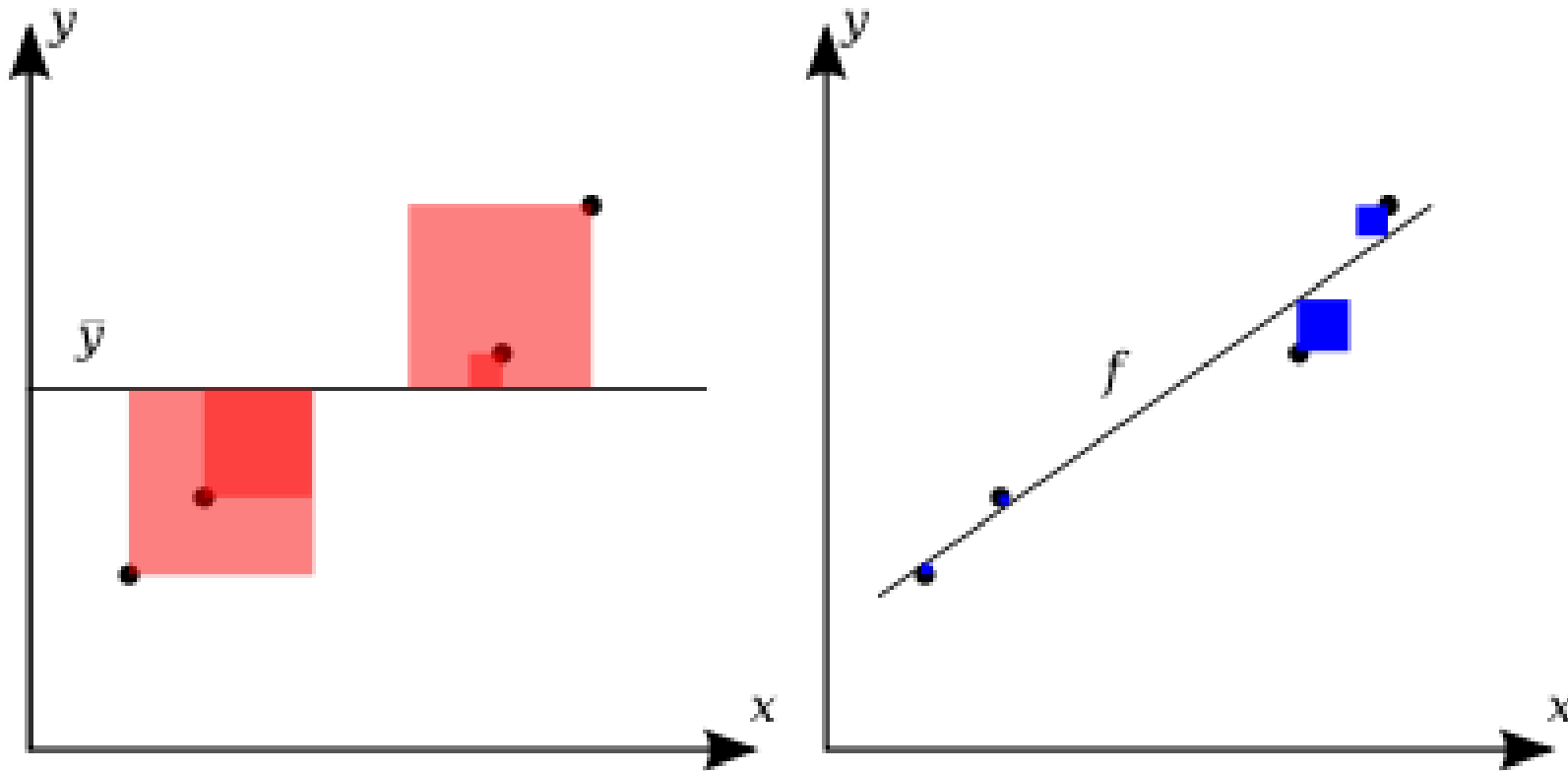
R-SQUARED (R2)

statistical measure that represents the **proportion of the variance** for a dependent variable that's explained by an independent variable or variables in a regression model

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$



R-SQUARED (R²)



$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Image from:
https://en.wikipedia.org/wiki/Coefficient_of_determination#/media/File:Coefficient_of_Determination.svg

R-SQUARED (R^2)

Importance

- **Goodness of Fit:** R^2 is a key indicator of the goodness of fit of a model. It shows how well the model explains the variability in the dependent variable.
- **Model Comparison:** R^2 can be used to compare the explanatory power of different models for the same dataset.

Limitations

- **Does Not Indicate Causation:** A high R^2 value does not imply that the model variables are causally related to the dependent variable.
- **Not Suitable for Non-Linear Models:** R^2 is primarily used for linear regression models and may not be appropriate for evaluating non-linear models.

MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

expresses the average absolute error
between the predicted and actual values
as a percentage of the actual values

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

Importance

- **Interpretability:** MAPE is **easy to understand and interpret**, as it represents the error as a percentage of the actual values.
- **Normalization:** MAPE provides a normalized measure of error, making it useful for comparing model performance across different datasets and scales.

Limitations

- **Division by Zero:** MAPE is undefined if any actual value (y_i) is zero, as this would involve division by zero.
- **Sensitivity to Small Values:** MAPE can be excessively high if the actual values are very small, as small differences between predicted and actual values result in large percentage errors.



03

TRAIN-TEST SPLIT AND CROSS VALIDATION

INTRODUCTION TO TRAIN-TEST SPLIT

Models should be robust,
generalizable, and capable of
performing well on unseen data

TRAIN-TEST SPLIT

Data set

RANDOM SPLIT



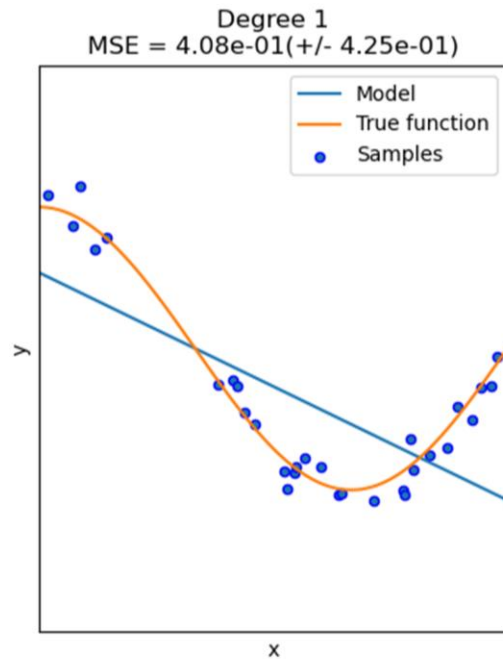
Training set

Test set

STRATIFIED SPLIT



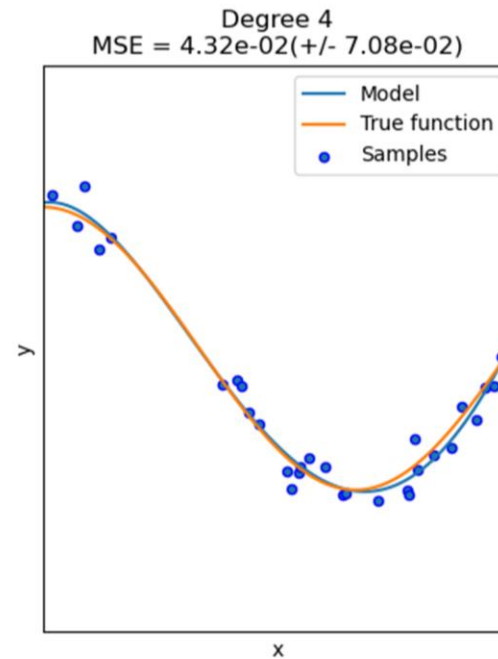
OVERFITTING AND UNDERFITTING



Training set



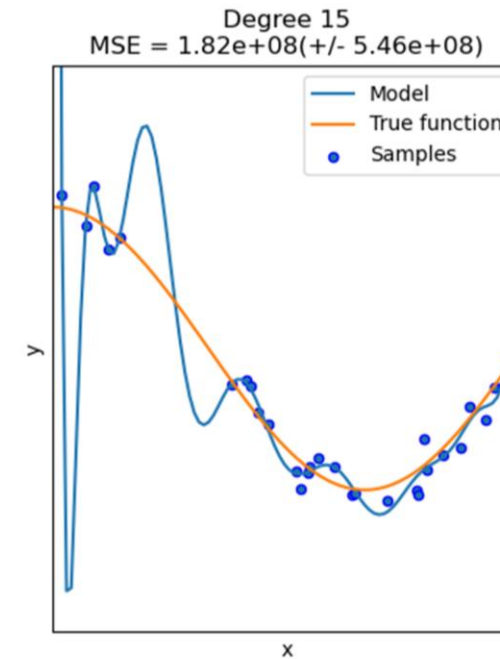
Test set



Training set



Test set



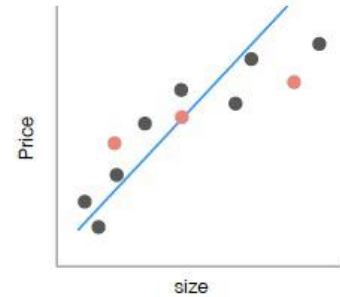
Training set



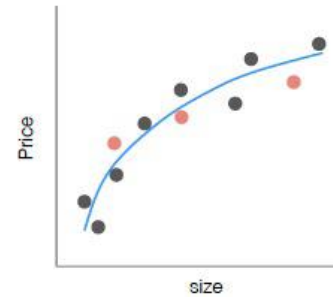
Test set

OVERFITTING AND UNDERFITTING

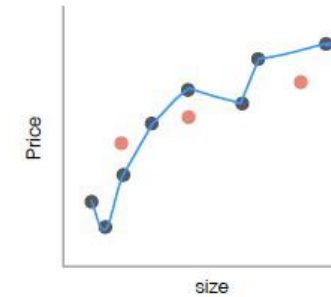
Regression
example



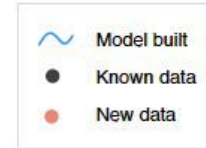
Under fitting model
(high training error, high test error)



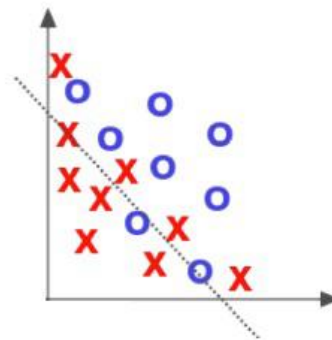
Good Fitting
(low training error, low test error)



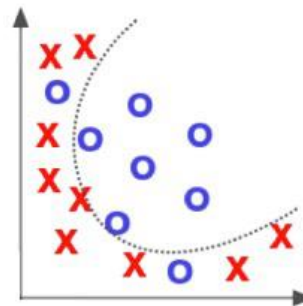
Overfitting Model
(no training error, high test error)



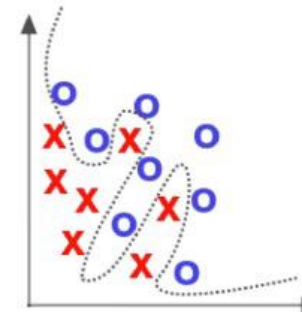
Classification
example



Under fitting model
(high training error, high test error)

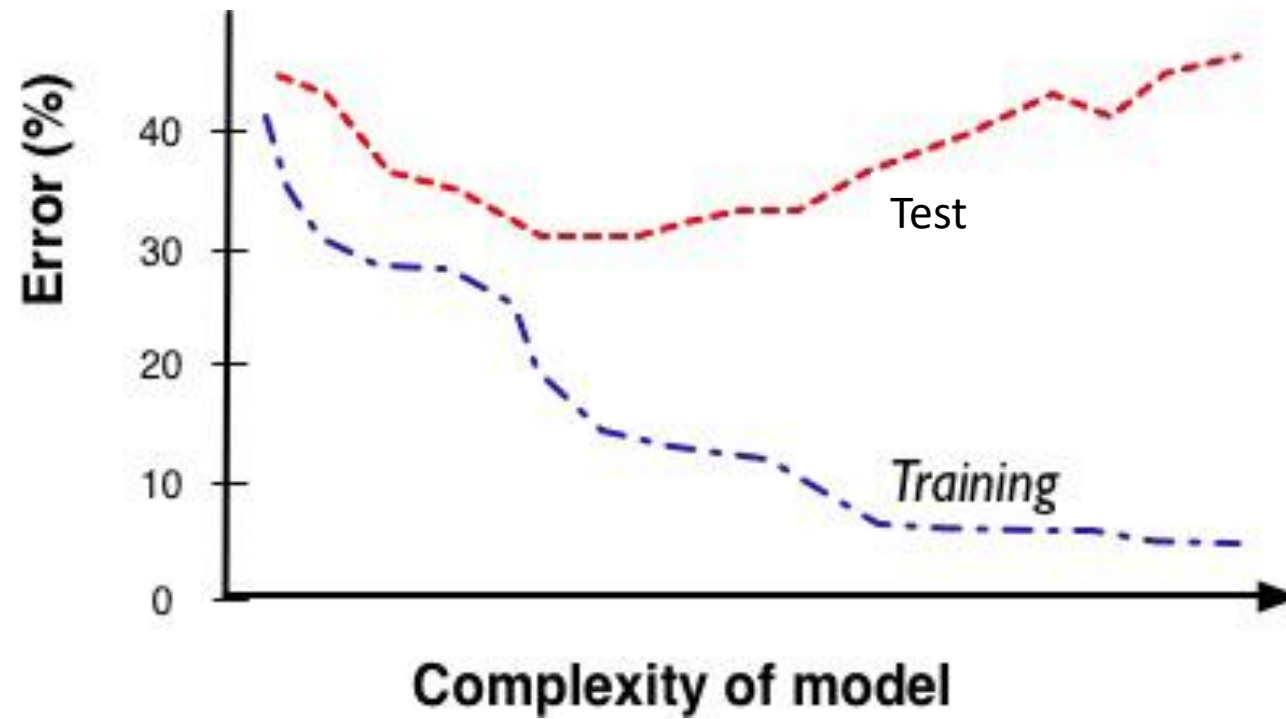


Good Fitting
(low training error, low test error)

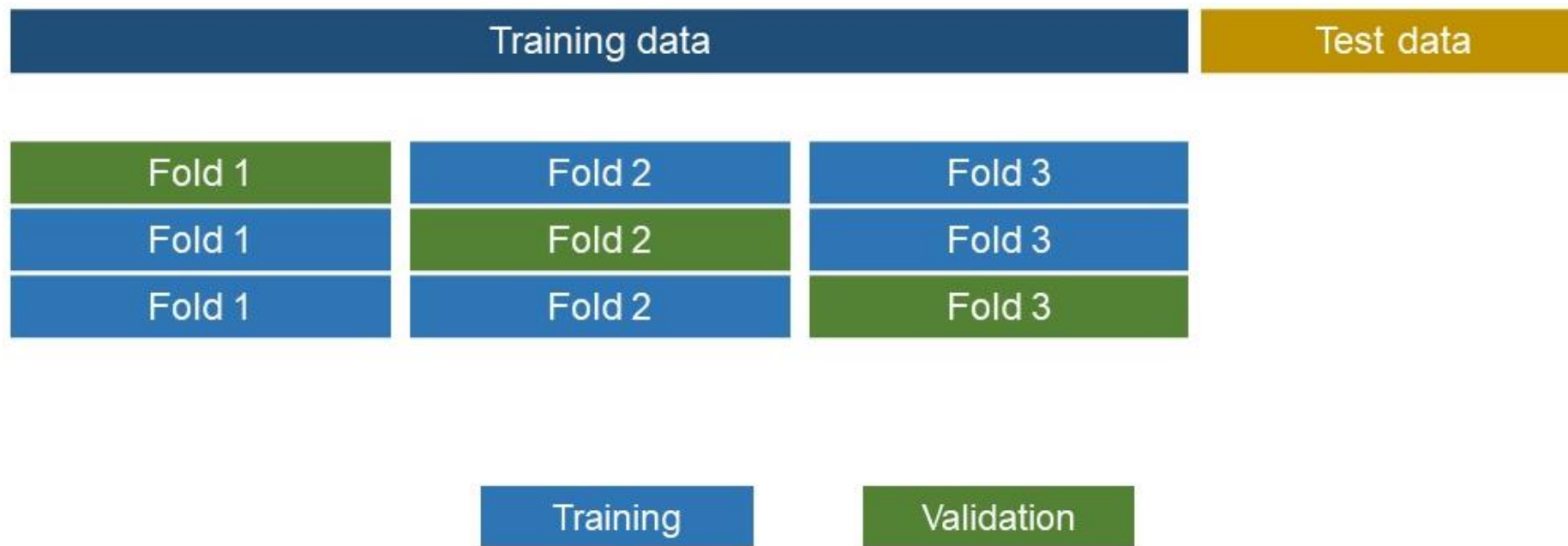


Overfitting Model
(no training error, high test error)

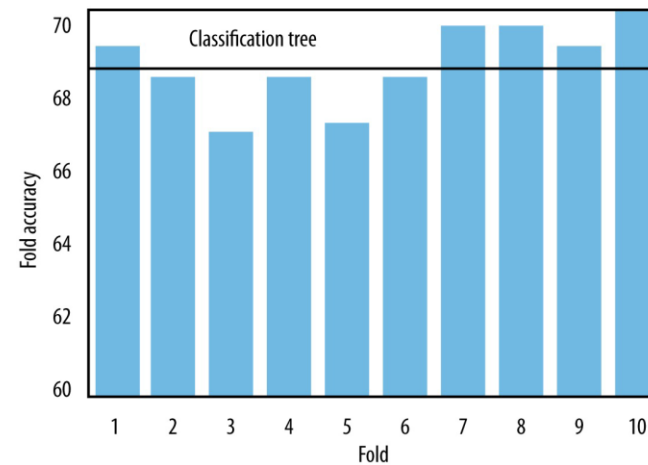
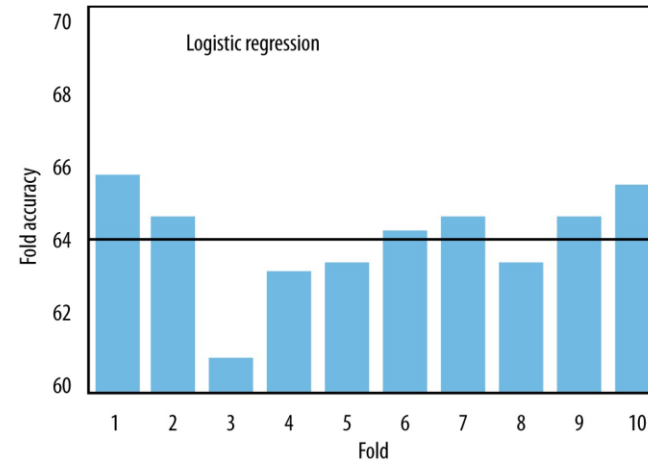
OVERFITTING AND UNDERFITTING



K-FOLD CROSS-VALIDATION

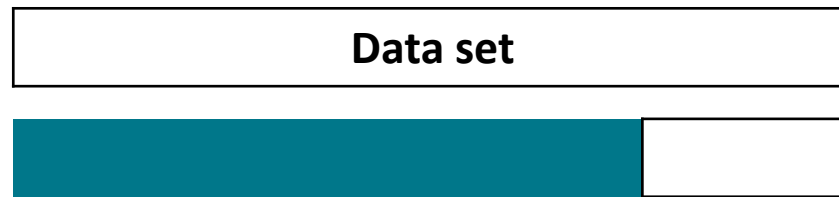


K-FOLD CROSS-VALIDATION



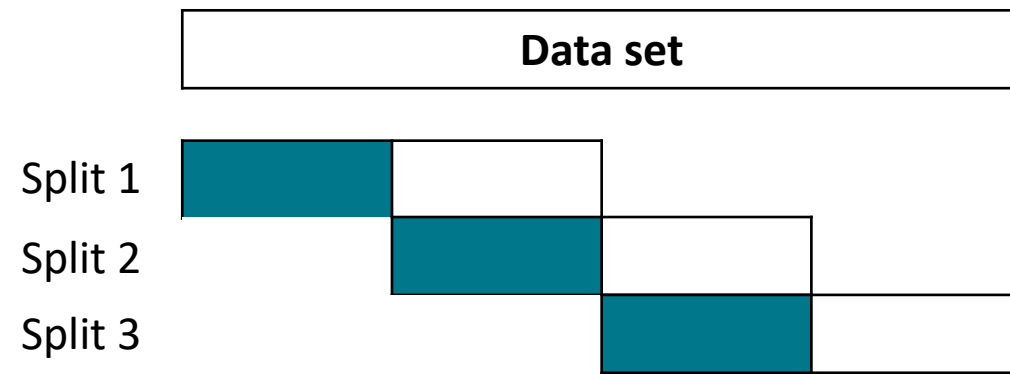
TIME-SERIES SPLIT

(for medical data with temporal component)



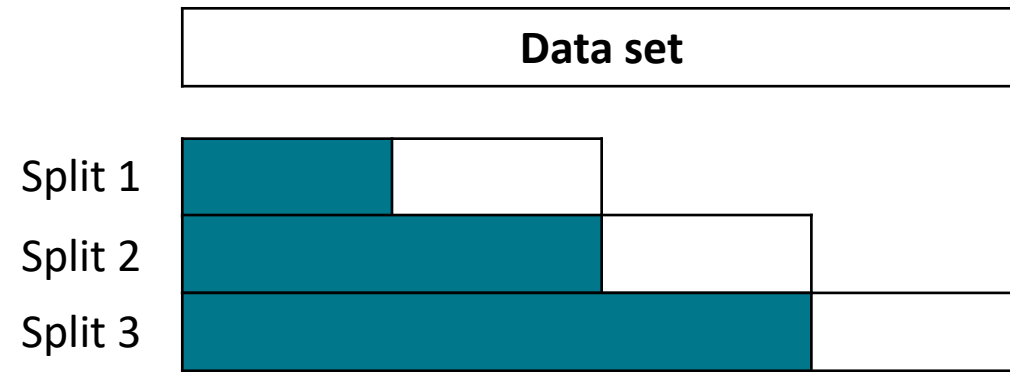
TIME-SERIES SPLIT – SLIDING WINDOW

(for medical data with temporal component)



TIME-SERIES SPLIT – EXPANDING WINDOW

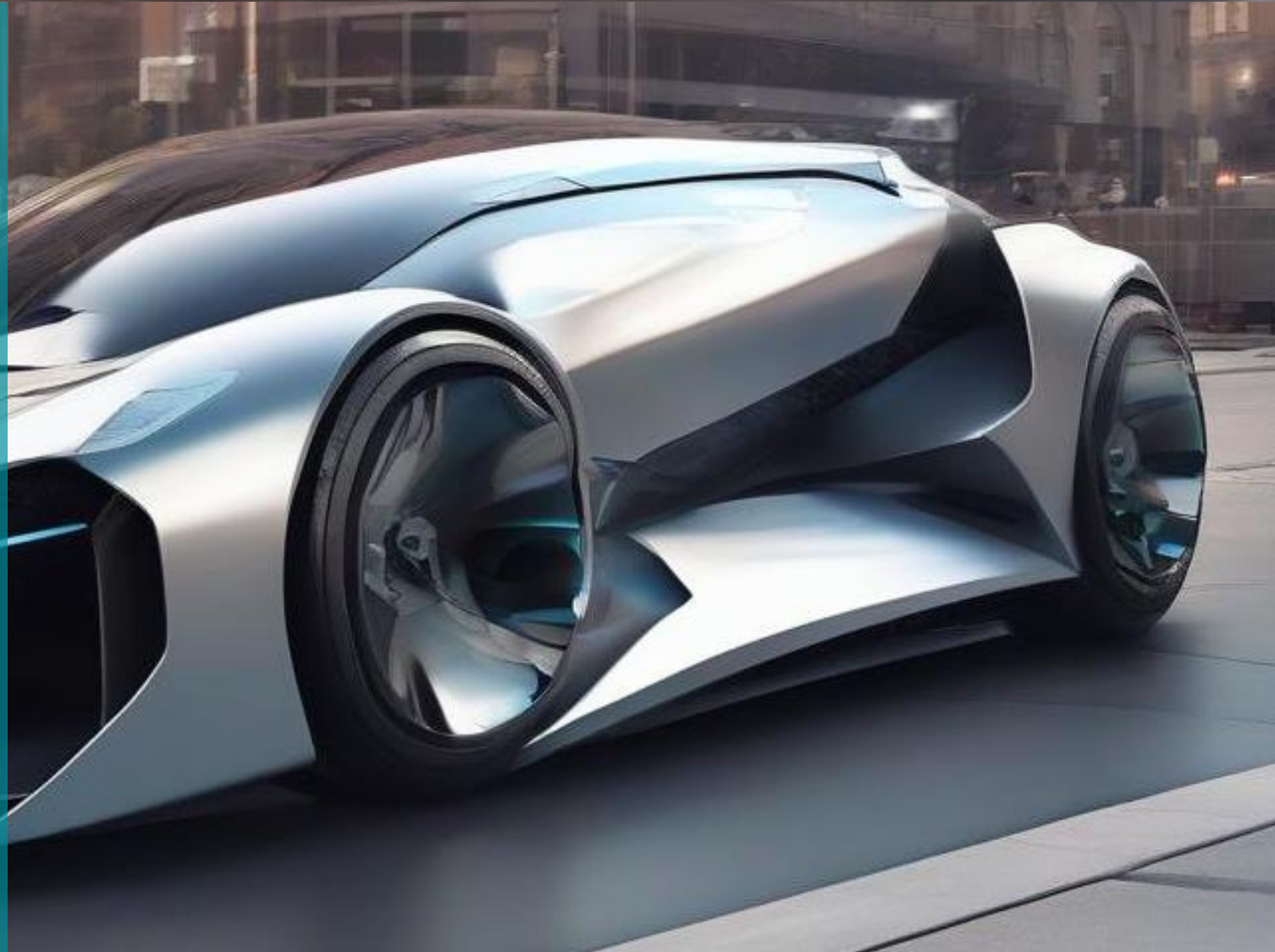
(for medical data with temporal component)



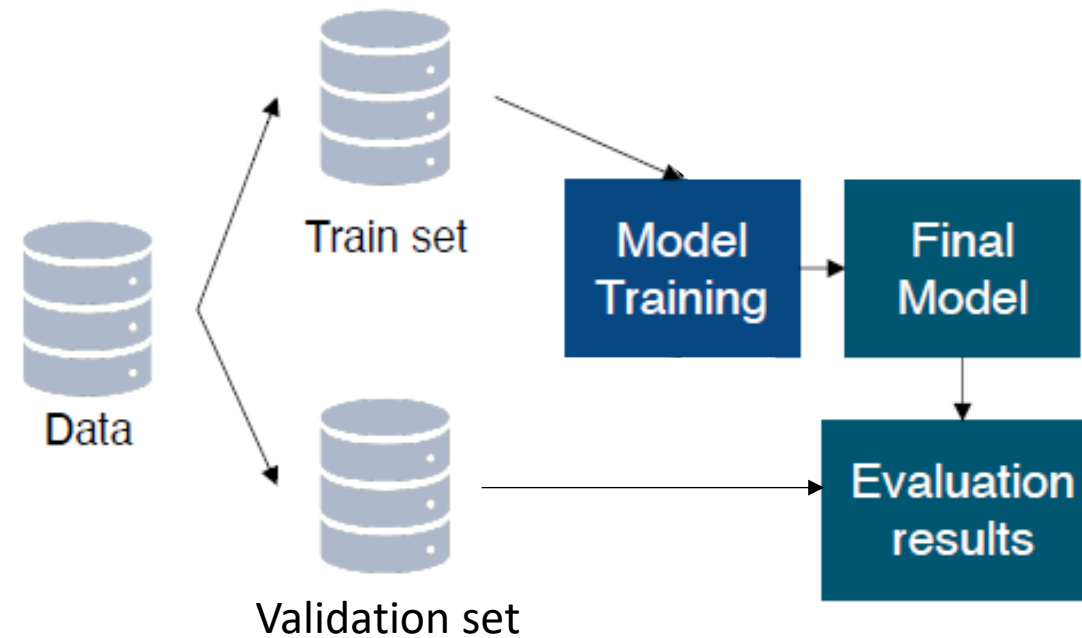
EXERCISE

04

MODEL TUNING AND OPTIMIZATION



INTRODUCTION TO MODEL TUNING



HYPERPARAMETERS AND PARAMETERS

Parameters	Hyperparameters
• Estimated during the training • Its part of the model • The estimated value is saved with the trained model to make prediction Model Training	• Values are set beforehand • External to the model • Not a part of the trained model and hence the vales are not saved for making prediction Hyperparameter tuning

HYPERPARAMETERS AND PARAMETERS

LOGISTIC REGRESSION

$$\hat{y} = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n)$$

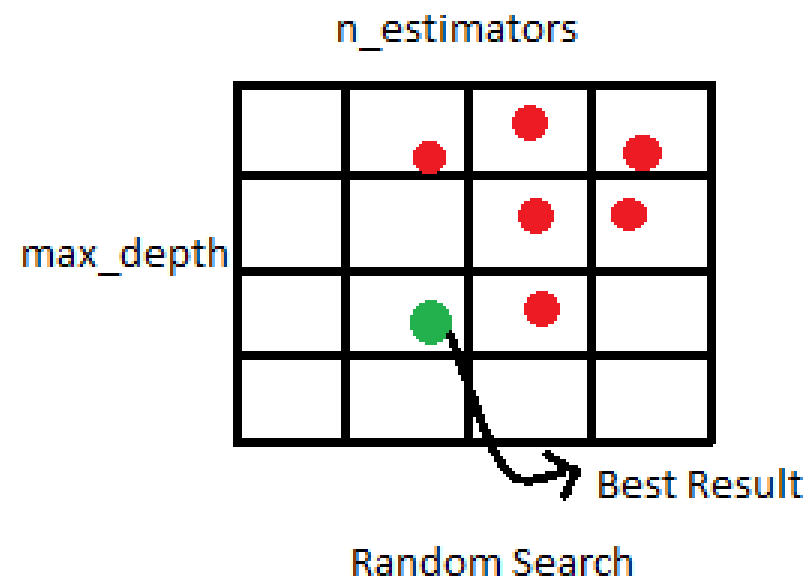
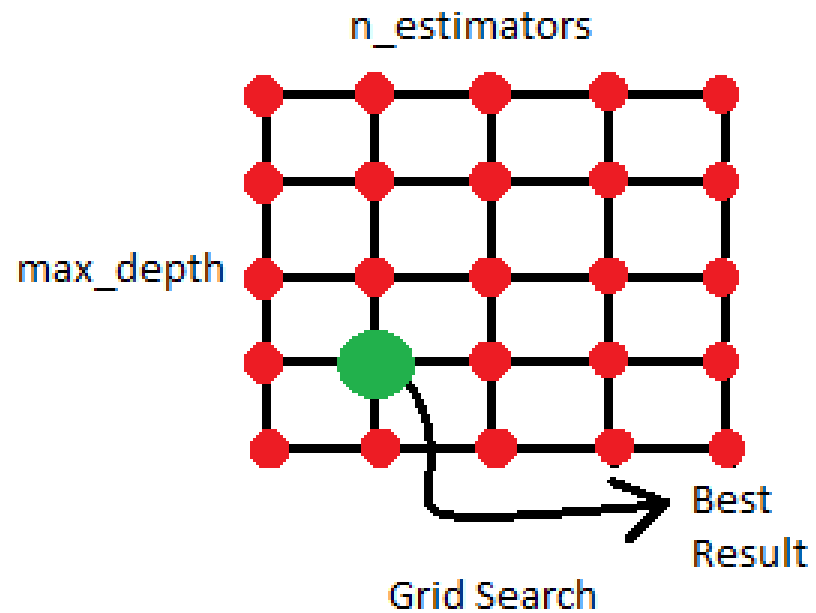
```
class sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False,  
tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None,  
random_state=None, solver='lbfgs', max_iter=100, multi_class='deprecated', verbose=0,  
warm_start=False, n_jobs=None, l1_ratio=None) \[source\]
```

HYPERPARAMETERS AND PARAMETERS

DECISION TREE

```
# Create a DecisionTreeClassifier object
dt_classifier = DecisionTreeClassifier(
    criterion='gini',          # Split criterion
    max_depth=5,              # Maximum depth of the tree
    min_samples_split=10,     # Minimum number of samples required to split an internal node
    min_samples_leaf=5,       # Minimum number of samples required to be at a leaf node
    max_features='sqrt',      # Number of features to consider when looking for the best split
    max_leaf_nodes=20,        # Maximum number of leaf nodes
    min_impurity_decrease=0.01, # Minimum impurity decrease required to split a node
    random_state=42           # Random state for reproducibility
)
```

GRID SEARCH VS. RANDOM SEARCH



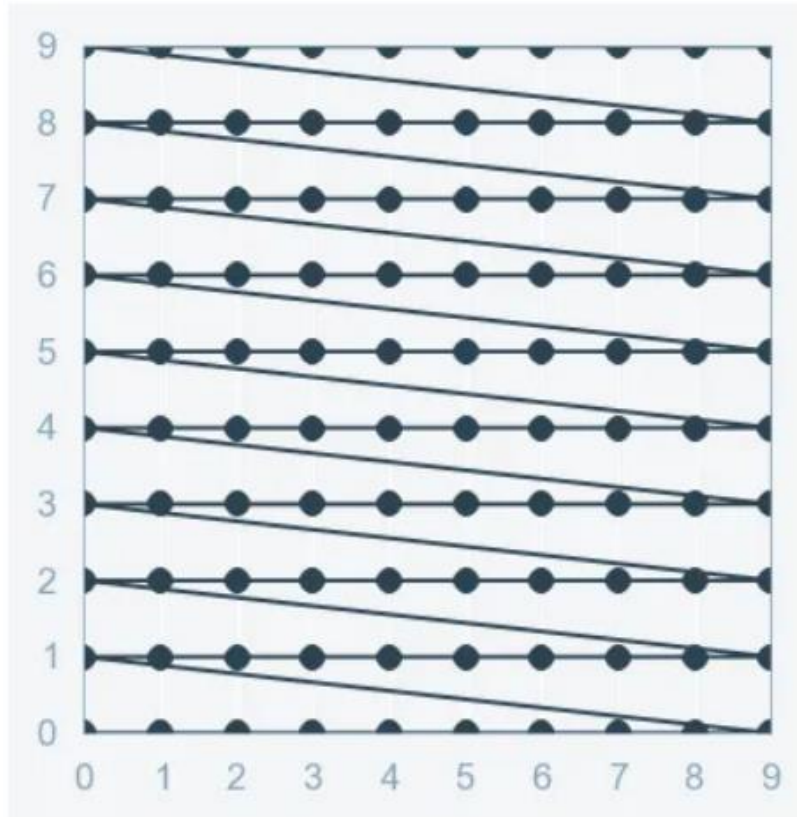
GRID SEARCH VS. RANDOM SEARCH

C	0.5	0.701	0.703	0.697	0.696
	0.4	0.699	0.702	0.698	0.702
	0.3	0.721	0.726	0.713	0.703
	0.2	0.706	0.705	0.704	0.701
	0.1	0.698	0.692	0.688	0.675
		0.1	0.2	0.3	0.4

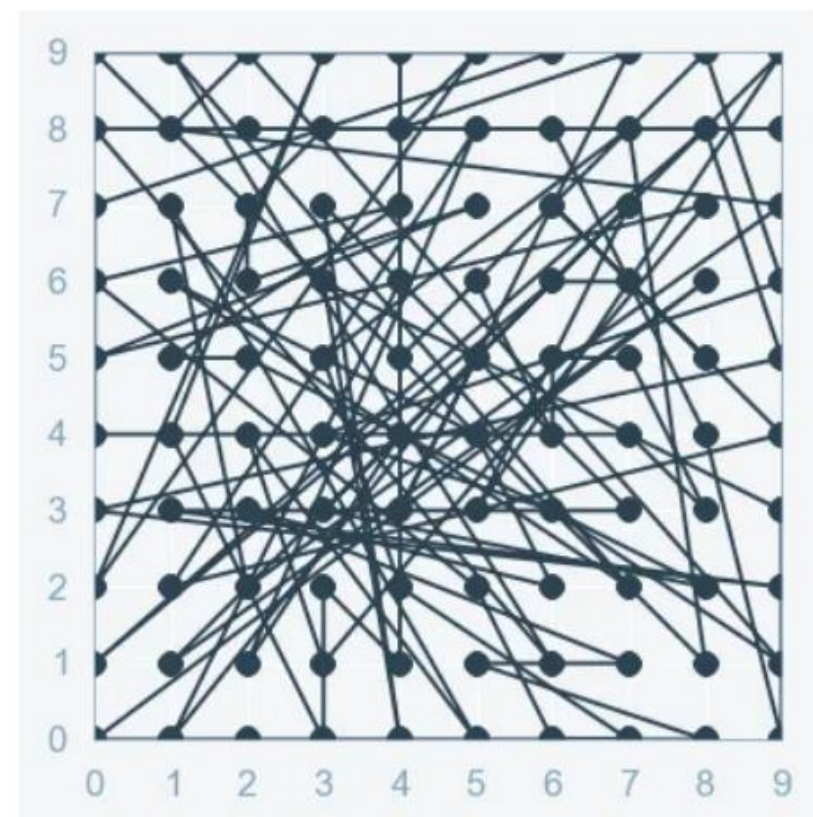
Alpha

GRID SEARCH VS. RANDOM SEARCH

GRID SEARCH



RANDOM SEARCH



EXERCISE

- GRID SEARCH

BREAK

GRADIENT DESCENT UPDATE RULE

LOGISTIC REGRESSION

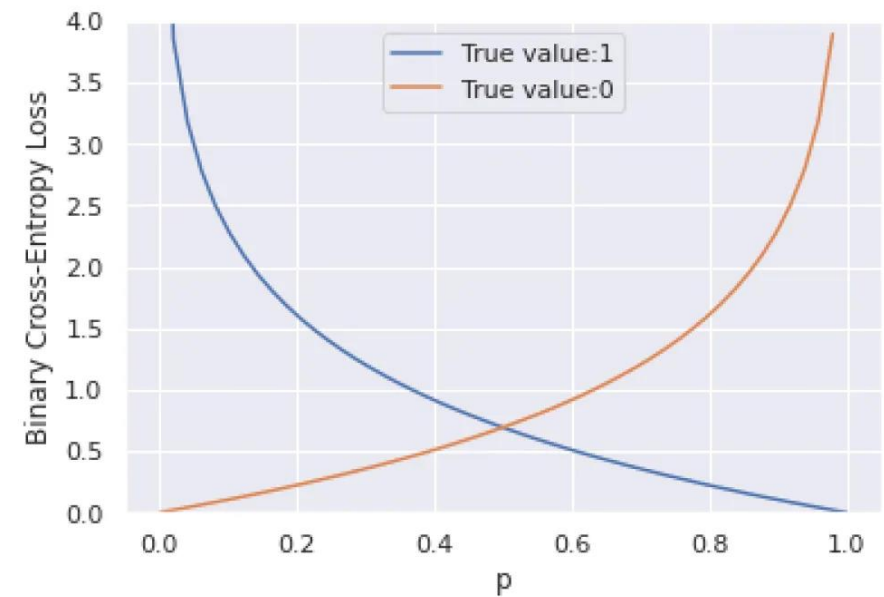
$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$

$$\hat{y} = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$$

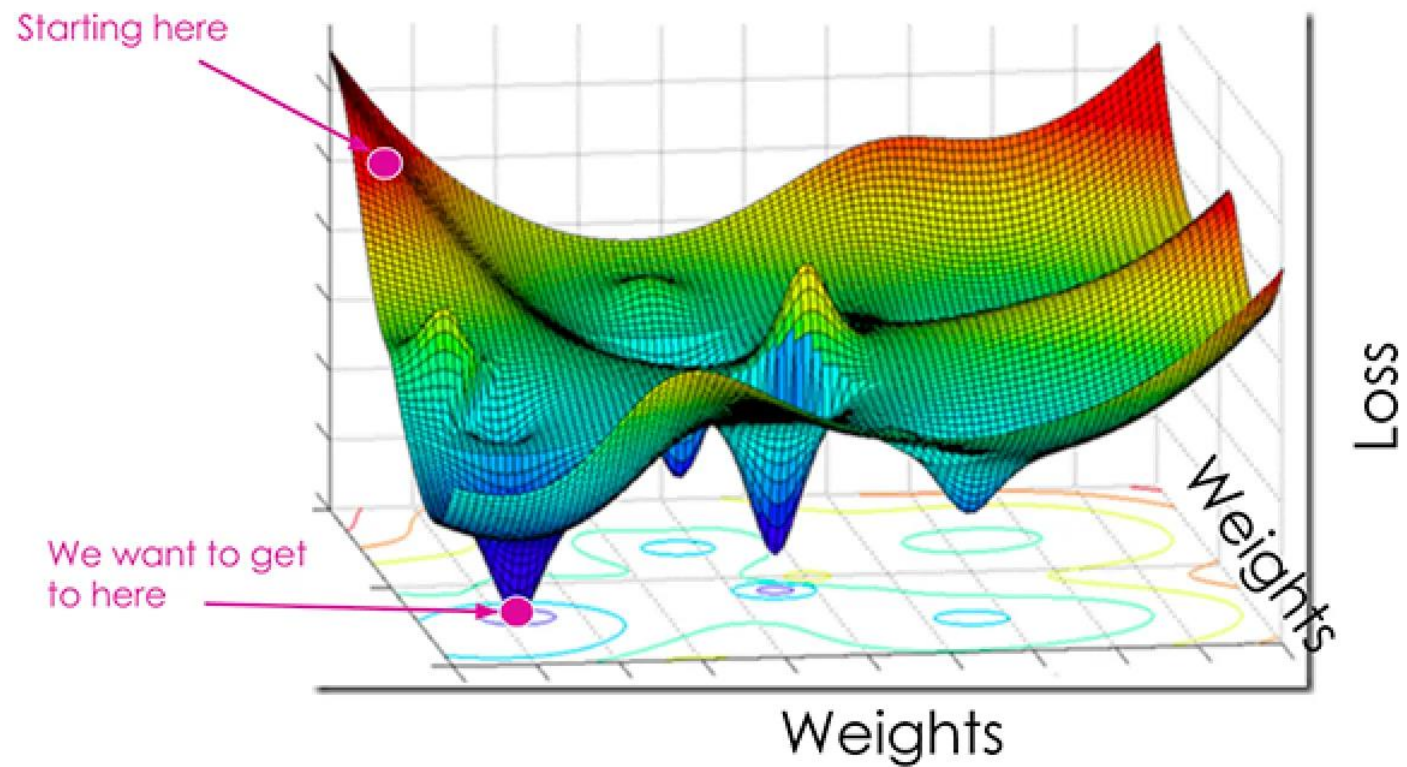
LOG LOSS

$$\text{Log Loss} = -\frac{1}{m} \sum_{i=1}^m [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$



OPTIMIZATION

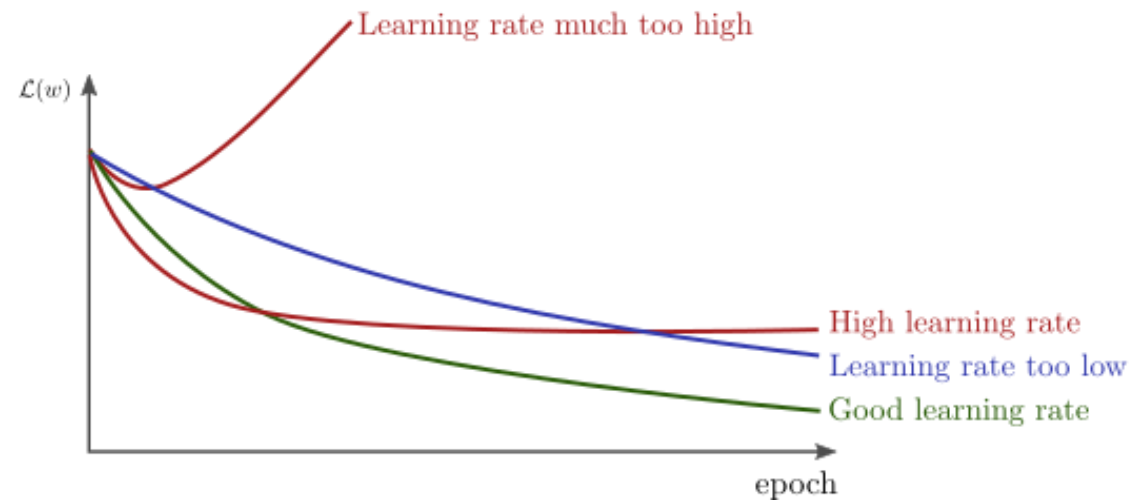
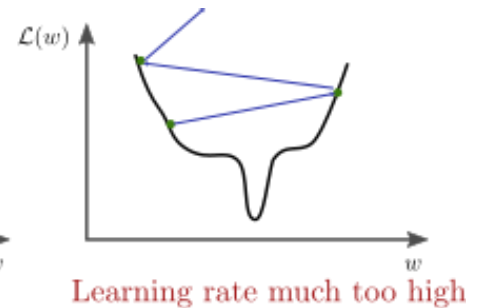
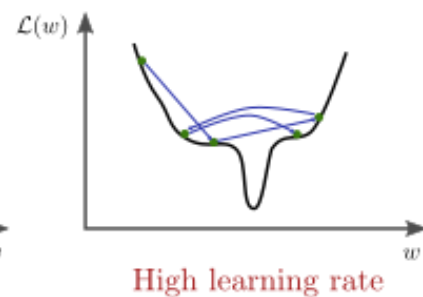
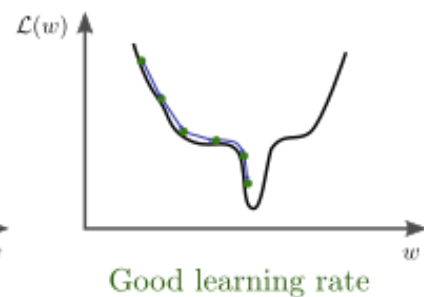
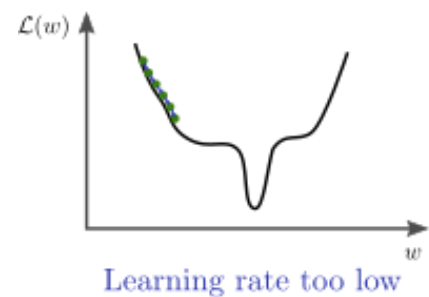


EXERCISE

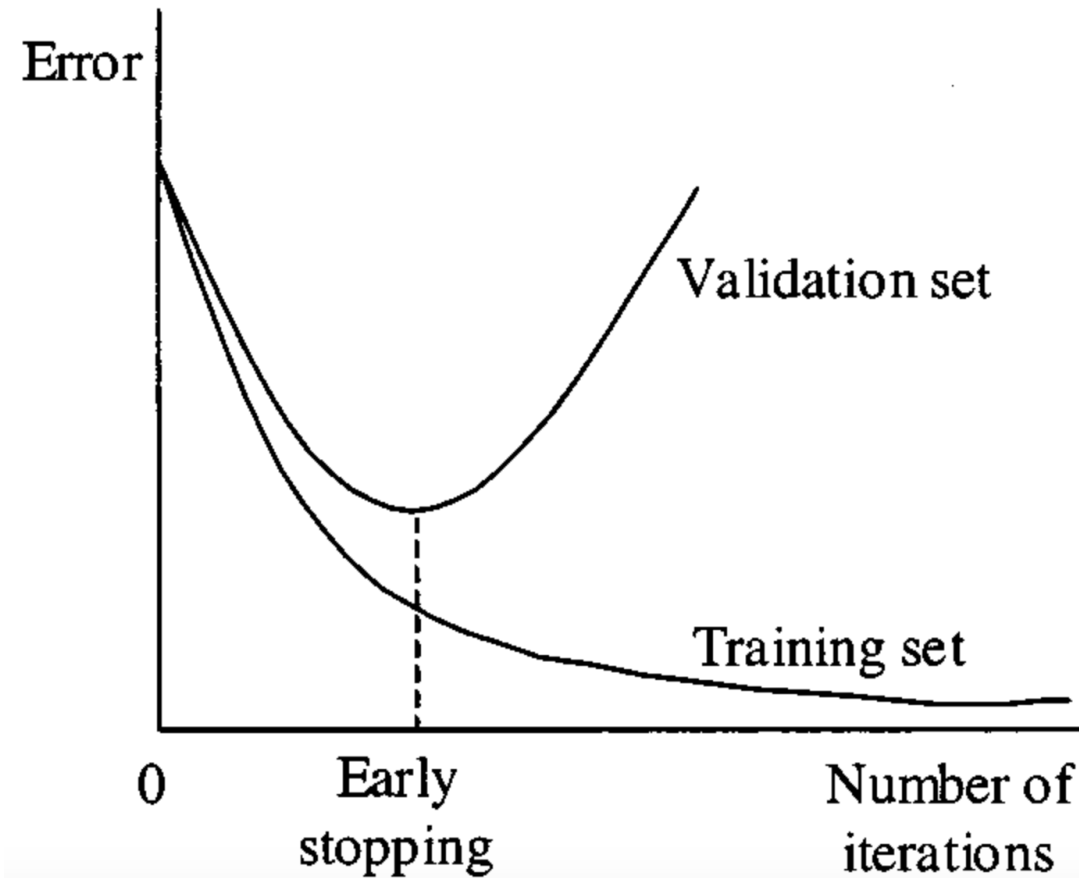
- GRADIENT DESCENT

LEARNING RATE

$$\beta_j = \beta_j - \alpha \frac{\partial \text{Log Loss}}{\partial \beta_j}$$

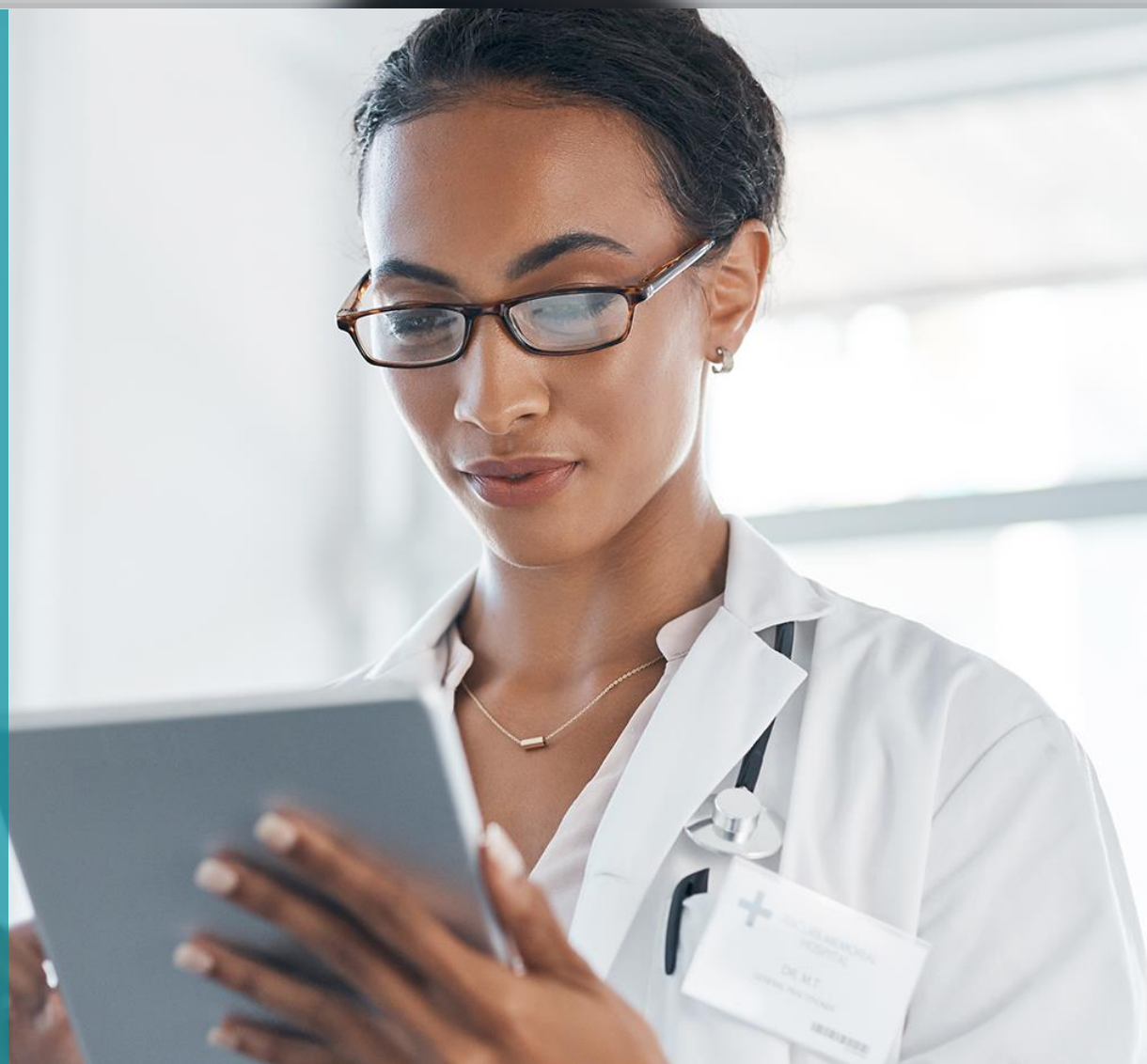



EARLY STOPPING

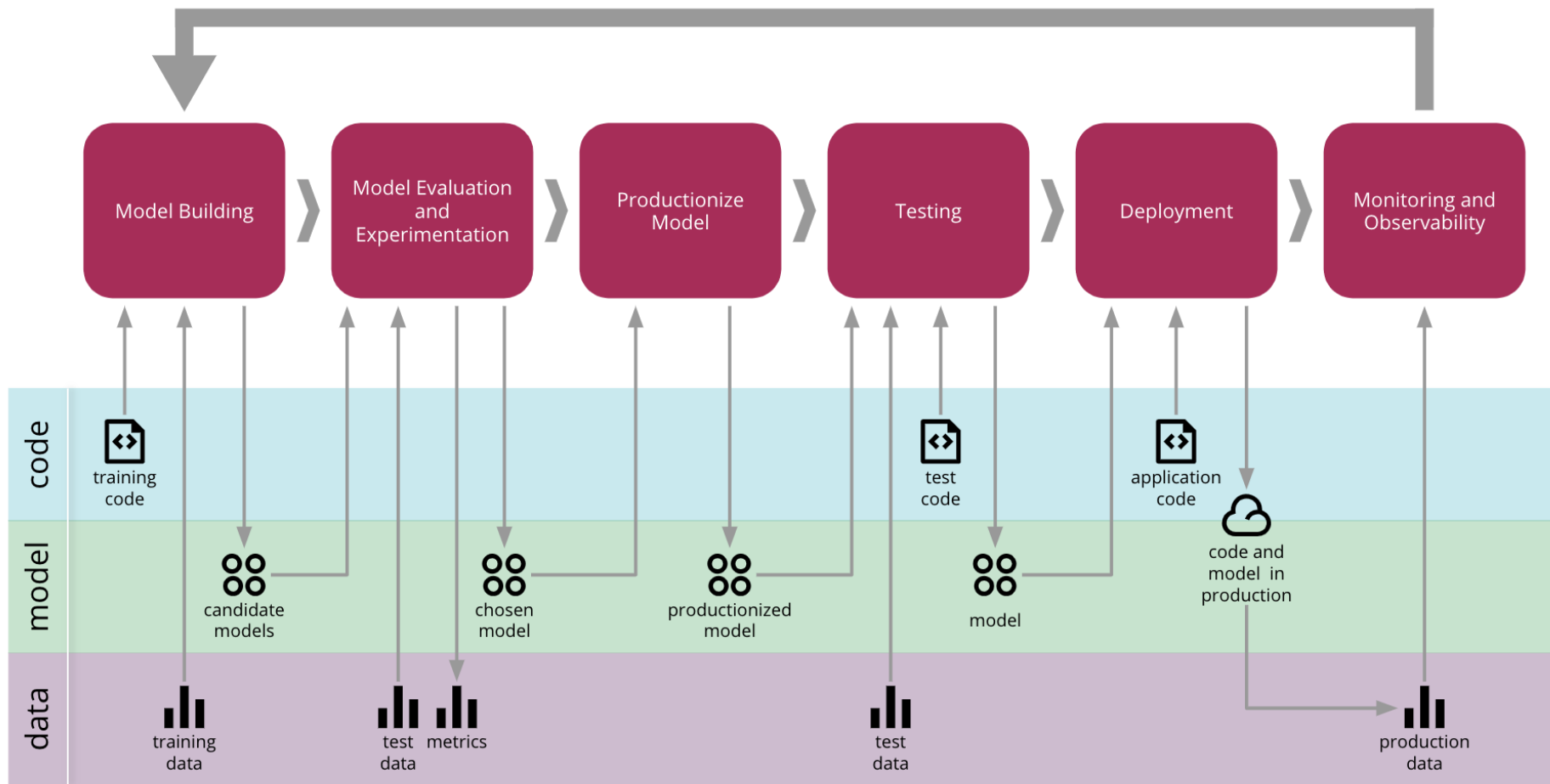


From <https://paperswithcode.com/method/early-stopping>

CONCLUSION



MODEL MONITORING



From <https://martinfowler.com/articles/cd4ml.html>

FURTHER READING

THANK YOU

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